New zero-density bounds near $\sigma = 0.77$ from the second Heath-Brown large values theorem

A note for the ANTEDB (Analytic Number Theory Exponent Database)

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Abstract

Combining the second Heath-Brown large values theorem (Lemma 7.15 of the blueprint) with Huxley subdivision (Lemma 7.4(ii)) *beyond* the end of the Montgomery range, and feeding the result into Corollary 11.9, we obtain the zero-density bounds

$$\mathrm{A}(\sigma) \le \frac{39}{301\sigma - 211}, \qquad \mathrm{A}(\sigma) \le \frac{15}{89\sigma - 61}, \qquad \mathrm{A}(\sigma) \le \frac{7}{29\sigma - 19},$$

which improve the current Table 11.1 strictly on the intervals (547/720, 127/167), (29/38, 13/17) and (107/139, 41/53) respectively (roughly $[0.7597, 0.7605]$, $[0.7632, 0.7647]$, $[0.7698, 0.7736]$). The gain is small (at most ≈ 0.04, e.g. 2.0486 instead of 2.0899 at $\sigma = 0.773$), but the argument uses only results already in the database, and every inequality has been verified in exact arithmetic. We also record a [Kerr-conditional] bound $\mathrm{A}(\sigma) \le 12/(37\sigma - 23)$ on $(37/47, 23/29)$, and a few small issues found in the blueprint and the Python code.

1 Statement

Theorem 1. *One has*

$$\begin{aligned} \mathrm{A}(\sigma) &\le \frac{39}{301\sigma - 211} && \text{for } \tfrac{241}{321} \le \sigma \le \tfrac{127}{167}, \\ \mathrm{A}(\sigma) &\le \frac{15}{89\sigma - 61} && \text{for } \tfrac{73}{97} \le \sigma \le \tfrac{13}{17}, \\ \mathrm{A}(\sigma) &\le \frac{7}{29\sigma - 19} && \text{for } \tfrac{25}{33} \le \sigma \le \tfrac{41}{53}. \end{aligned}$$

These bounds are strictly better than Table 11.1 exactly on (547/720, 127/167)*,* (29/38, 13/17) *and* (107/139, 41/53) *respectively, and coincide with the table at the right endpoints* 127/167*,* 13/17*,* 41/53*.*

The three bounds are the cases $k = 5, 4, 3$ of the following family.

Proposition 2. *Let $k \ge 1$ be an integer and $3/4 < \sigma < 1$. Put*

$$\tau' := \frac{2k(3 - 2\sigma)}{3k - 2}, \qquad h := \frac{4k\big(3k - 1 - (4k - 2)\sigma\big)}{3k - 2},$$

*Computations and drafting assisted by Claude (Anthropic). All numbered references (Lemma 7.15, Corollary 11.9, Table 11.1, ...) are to the ANTEDB blueprint, version of 26 September 2026.

$$\tau_0 := 3-3\sigma+\tau'-h = \frac{(16k^2-21k+6)\sigma-(12k^2-19k+6)}{3k-2}.$$

Suppose that

$$\begin{gathered} h \ge 2-2\sigma, \qquad 1 \le \tau' \le \tau_0 \le \tfrac{3}{2}, \qquad \tau_0 \ge 3-3\sigma, \\ k+k(2-4\sigma) \le \frac{3-3\sigma}{\tau_0}, \qquad \tfrac{2}{3}+\tfrac{k(12-16\sigma)}{3} \le \frac{3-3\sigma}{\tau_0}, \qquad h \le \frac{(3-3\sigma)\tau'}{\tau_0}. \end{gathered} \tag{1}$$

Then

$$\mathrm{A}(\sigma) \le \frac{3}{\tau_0} = \frac{3(3k-2)}{(16k^2-21k+6)\sigma-(12k^2-19k+6)}.$$

For $k=3,4,5$ the conditions (1) hold exactly on the intervals stated in Theorem 1 (see §8). For instance, the condition $h \ge 2-2\sigma$ is equivalent to $\sigma \le \frac{6k^2-5k+2}{8k^2-7k+2}$ (giving the right endpoints 41/53, 13/17, 127/167; this is exactly the first threshold in Lemma 11.30), and $\tau' \le \tau_0$ is equivalent to $\sigma \ge \frac{12k^2-13k+6}{16k^2-17k+6}$ (giving the left endpoints 25/33, 73/97, 241/321).

2 Inputs

Only the following results of the blueprint are used.

- **Theorem 7.9** (L^2 mean value theorem): $\mathrm{LV}(\sigma,\tau) \le \max(2-2\sigma,\, 1+\tau-2\sigma)$.
- **Lemma 7.15** (second Heath-Brown large values theorem): for $3/4 < \sigma \le 1$, $\tau \ge 0$ and any positive integer k,

  $$\mathrm{LV}(\sigma,\tau) \le \max\bigl(2-2\sigma,\; k\tau+k(2-4\sigma),\; \tfrac{2\tau}{3}+\tfrac{k(12-16\sigma)}{3}\bigr).$$

  (The same bound appears, with m in place of k, as the X-term of Lemma 11.29.)
- **Lemma 7.4(ii)** (Huxley subdivision): $\mathrm{LV}(\sigma,\tau_2) \le \mathrm{LV}(\sigma,\tau_1)+\tau_2-\tau_1$ for $\tau_2 \ge \tau_1$.
- **Corollary 11.9**: if $\mathrm{LV}(\sigma,\tau) \le (3-3\sigma)\tau/\tau_0$ for $2\tau_0/3 \le \tau \le \tau_0$ (11.6) and $\mathrm{LV}_\zeta(\sigma,\tau) \le (3-3\sigma)\tau/\tau_0$ for $2 \le \tau < 4\tau_0/3$ (11.7), then $\mathrm{A}(\sigma) \le 3/\tau_0$.

3 Proof of Proposition 2

Write $M := 2-2\sigma$, $L_1(\tau) := k\tau+k(2-4\sigma)$, $L_2(\tau) := \frac{2\tau}{3}+\frac{k(12-16\sigma)}{3}$ and $\rho(\tau) := (3-3\sigma)\tau/\tau_0$ (the "ray"). One checks directly that $L_1(\tau') = L_2(\tau') = h$, i.e. τ' is the point where the two non-Montgomery terms of Lemma 7.15 cross, and that $h+(\tau_0-\tau') = 3-3\sigma = \rho(\tau_0)$.

Hypothesis (11.7). Since $\tau_0 \le 3/2$, the range $2 \le \tau < 4\tau_0/3$ is empty, so (11.7) holds vacuously. (Empty ranges are permitted: the first supremum in Corollary 11.7 is $-\infty$ in that case; the blueprint uses the same observation in the proofs of Theorems 11.11, 11.15 and 11.16.)

Hypothesis (11.6). Since $\tau_0 \le 3/2$ we have $2\tau_0/3 \le 1 \le \tau' \le \tau_0$, and we split $[2\tau_0/3, \tau_0]$ into three pieces.

(a) $2\tau_0/3 \le \tau \le 1$. By Theorem 7.9, $\mathrm{LV}(\sigma,\tau) \le \max(M, 1+\tau-2\sigma) = M$, and $M \le \rho(\tau)$ for every $\tau \ge 2\tau_0/3$ (this is (11.8)).

(b) $1 \le \tau \le \tau'$. By Lemma 7.15 (valid as $\sigma > 3/4$), $\mathrm{LV}(\sigma,\tau) \le \max(M, L_1(\tau), L_2(\tau))$. We have $M \le \rho(\tau)$ as in (a). The functions L_1, L_2, ρ are affine in τ, so it suffices to check $L_i \le \rho$ at $\tau = 1$

and at $\tau = \tau'$. At $\tau = 1$ these are two of the conditions (1); at $\tau = \tau'$ both equal h, and $h \le \rho(\tau')$ is the last condition in (1).

(c) $\tau' \le \tau \le \tau_0$. By Lemma 7.4(ii), then (b) at $\tau = \tau'$ and $h \ge M$,

$$\mathrm{LV}(\sigma, \tau) \le \mathrm{LV}(\sigma, \tau') + \tau - \tau' \le h + \tau - \tau' =: H(\tau).$$

H is affine with slope 1, $H(\tau_0) = 3-3\sigma = \rho(\tau_0)$, and ρ has slope $(3-3\sigma)/\tau_0 \le 1$ because $\tau_0 \ge 3-3\sigma$. Hence $H(\tau) \le \rho(\tau)$ for $\tau \le \tau_0$.

Thus (11.6) holds on $[2\tau_0/3, \tau_0]$, and Corollary 11.9 gives $\mathrm{A}(\sigma) \le 3/\tau_0$. □

Remark 3 (What is new compared with Ivić). Lemma 11.30 / Corollary 11.31 use the same Heath-Brown bound only on the range where it equals the Montgomery value $2-2\sigma$, and then extend with slope 1 via Lemma 7.7. This requires the crossing value h to satisfy $h \le 2-2\sigma$, i.e. $\sigma \ge \frac{6k^2-5k+2}{8k^2-7k+2}$. For σ slightly below this threshold, the slope-2/3 term L_2 leaves the Montgomery value *before* the slope-k term does. Following L_2 (slope $2/3 < 1$) up to its crossing τ' with L_1, and only then switching to slope-1 subdivision, gives a larger admissible τ_0. At the threshold the two arguments coincide, which is why each new bound meets Ivić's bound $3k/((3k-2)\sigma + 2 - k)$ at the right endpoint of its interval.

Remark 4 (Optimality within the framework). The admissible τ_0 is exactly the solution of $\mathrm{LV}(\sigma, \tau_0) = 3-3\sigma$ for the upper bound on LV used here, i.e. the limit predicted by Heuristic 11.38. An independent exhaustive piecewise-linear search over all large-value estimates in the blueprint (closed under subdivision, monotonicity and raising to powers, combined with Corollaries 11.7–11.10) reproduces Table 11.1 exactly on $[0.76, 0.7955]$ and improves on it only on the three intervals of Theorem 1, where it agrees with the formulas above. A re-optimization of the Bourgain-type argument (Theorem 11.22, Corollaries 10.30/10.31) gave no further improvement on $[0.76, 0.80]$.

4 Comparison with Table 11.1

σ	new bound	Table 11.1	table entry
0.7600	2.1959	2.2059	$9/(8\sigma-2)$ (Cor. 11.31)
0.7603	2.1848	2.2046	$9/(8\sigma-2)$
0.7640	2.1441	2.1639	$15/(13\sigma-3)$ (Cor. 11.31)
0.7710	2.0840	2.1016	$6/(5\sigma-1)$ (Cor. 11.31)
0.7720	2.0661	2.0979	$6/(5\sigma-1)$
0.7730	2.0486	2.0899	$2/(9\sigma-6)$ (Thm. 11.22)

Suggested new rows of Table 11.1 (with the neighbouring rows shortened accordingly):

bound	range
$39/(301\sigma-211)$	$547/720 \le \sigma \le 127/167$
$15/(89\sigma-61)$	$29/38 \le \sigma \le 13/17$
$7/(29\sigma-19)$	$107/139 \le \sigma \le 41/53$

Note that the first row also slightly improves the Guth–Maynard bound $15/(3+5\sigma)$ on $(547/720, 19/25)$. The bounds do not affect $\|\mathrm{A}\|_\infty$ (attained near $\sigma = 0.7$), hence do not change the prime-gap consequences of Proposition 15.4.

5 Why the database did not find this

`literature.py` contains no hypothesis encoding Lemma 7.15. Ivić's estimates enter only through the closed formula `ivic_ep_to_zd` (Lemma 11.30), which uses the Montgomery-range version of the Heath-Brown bound. Consequently the derived large-value region never sees the slope-2/3 piece. When Lemma 7.15 is added as a large-value hypothesis (for $k = 3, 4, 5$), together with the Huxley tail, the database's own routines `compute_large_value_region` and `compute_sup_rho_on_tau` on $[2\tau_0/3, \tau_0]$ reproduce exactly $7(1-\sigma)/(29\sigma-19)$, $15(1-\sigma)/(89\sigma-61)$ and $39(1-\sigma)/(301\sigma-211)$ for $\mathrm{A}(\sigma)(1-\sigma)$.

6 A Kerr-conditional bound

Proposition 5 (conditional on the unpublished preprint of Kerr)**.** *Assuming Lemma 10.32(iii), one has* $\mathrm{A}(\sigma) \le \frac{12}{37\sigma-23}$ *for* $\frac{37}{47} \le \sigma \le \frac{23}{29}$*. This is strictly better than* $9/(8(2\sigma-1))$ *(Theorem 11.22) on* $(37/47, 23/29)$*, by up to* ≈ 0.027*, and meets Kerr's bound* $3/(2\sigma)$ *(Proposition 11.37) at* $\sigma = 23/29$*.*

Sketch. Apply Corollary 11.9 with $\tau_0 = (37\sigma-23)/4$. For $\tau \le 3-2\sigma$, use the Montgomery range of Theorem 7.16 ($k=3$) followed by Huxley subdivision (unconditional). For $3-2\sigma \le \tau \le \tau_0$, use Lemma 10.32(iii) with $\alpha = (6\tau - 18 + 12\sigma)/5 \ge 0$ (strictly inside the admissible α-range on the open interval); the a priori bound $\mathrm{LV} \le \min(1, 4-2\tau)$ required there is supplied by the previous step. The resulting bound is the line $(4\tau + 38 - 52\sigma)/5$, which equals $3-3\sigma$ at τ_0 and has slope $4/5$, larger than that of the ray. Here $\tau_0 \ge 3/2$, so (11.7) is not vacuous; it follows from Theorem 9.7: $\mathrm{LV}_\zeta(\sigma,\tau) \le 2\tau - 12(\sigma - 1/2)$, which lies below the ray at $\tau = 2$ and at $\tau = 4\tau_0/3$, hence on $[2, 4\tau_0/3)$. All inequalities were verified symbolically on the whole interval. □

7 Small issues found along the way

1. **Proof of Lemma 7.15.** The displayed inequality $2(\rho+\sigma) \le \max(\rho+1, 2\rho+1/2, \dots)$ drops the trailing factor N of $(|W|V)^2 \ll T^{o(1)}(\cdots)N$; it should read $2(\rho+\sigma) \le 1+\max(\rho+1, 2\rho+1/2, \dots)$. The stated conclusion of the lemma is correct (it is what one gets with the factor N kept).

2. **Lemma 10.32(iii) (Kerr).** As transcribed, the admissible range of α allows $\alpha < 0$, and some negative values of α give $\mathrm{LV}(\sigma,\tau) < 2-2\sigma$, contradicting Lemma 7.5: for $\sigma = 0.79$, $\tau = 1.2$ the admissible range is $-0.44 < \alpha < 0.48$, and $\alpha = -0.17$ yields $\mathrm{LV}(\sigma,\tau) \le \max(0.25, 0.25, 0.14, 0.4067) < 0.42 = 2-2\sigma$. Probably a condition $\alpha \ge 0$ (as in part (i)) is missing. The Kerr-conditional bound above uses only $\alpha \ge 0$.

3. **Proof of Corollary 11.10.** "$2\tau_3 \le \tau$" should be "$2\tau_0/3 \le \tau$".

4. **Python code.** (a) `large_values.apply_huxley_subdivision` reads `original_region.polytopes`, but `Region` has no such attribute (it stores `children`). (b) In `functions.py`, `RationalFunction._compute_optimal` calls `sympy.real_roots` on non-polynomial expressions (e.g. involving square roots), which raises `PolynomialError`; this makes `derived.compute_best_zero_density()` and several `prove_zero_density_*` routines crash with current sympy. A fallback to `sympy.solveset(..., domain=S.Reals)` fixes it.

8 Verification

The accompanying script `verify_hb2_zero_density.py` (sympy only) checks, for $k = 3, 4, 5$ and in exact arithmetic (rational root isolation in σ, no sampling):

- the closed forms of τ', h, τ_0, $3/\tau_0$ and the identity $h + \tau_0 - \tau' = 3 - 3\sigma$;
- that all conditions (1) (and $\sigma > 3/4$) hold exactly on the intervals of Theorem 1, and that these are the maximal such intervals;
- that the new bound is strictly below every row of Table 11.1 exactly on the stated open intervals.

It prints `ALL OK`. In addition, the result was checked independently three times (an independent symbolic re-derivation, an audit of the statements against the LaTeX source of the blueprint, and an adversarial check at more than 2×10^5 exact rational points (σ, τ), plus a reproduction with the database's own polytope routines).

Caveats. The argument relies on Lemma 7.15 as stated in the blueprint (citing Heath-Brown, *Zero density estimates for the Riemann zeta-function and Dirichlet L-functions*, Lemma 6); the original paper was not re-checked. A brief literature search did not find these bounds, but novelty has not been established.